Medical Transcription NLP Classification

Project Based Assignment — Option 3

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Qualitative Analysis

Background / Motivation

Hospitals generate thousands of unstructured clinical documents daily — discharge summaries, operative notes, and consultation reports. Manually routing these records to the correct medical specialty is time-consuming and error-prone, consuming hours of senior physicians' time that could be better spent on direct patient care. The challenge is compounded by inconsistent terminology, heavy use of abbreviations (e.g., pt, hx, dx), negated findings ("No chest pain"), and overlapping vocabulary between specialties such as Orthopaedic and Neurology.

Natural Language Processing (NLP) offers a principled solution: a pipeline that cleans raw text, extracts clinical entities, handles negation, and automatically classifies each document into the appropriate specialty. This project addresses that problem using the publicly available MTSamples (Medical Transcriptions) dataset, demonstrating that an accuracy of nearly 95% is achievable with lightweight, interpretable machine-learning methods — no deep learning required.

Objectives

1. Build a robust text-preprocessing pipeline that removes clinical noise, expands medical abbreviations, tokenises, and lemmatises raw transcription text.
2. Extract domain-relevant entities (symptoms, body-part nouns, severity adjectives) using POS tagging and visualise specialty-specific vocabulary differences through word clouds.
3. Implement a syntactic negation-detection module that correctly identifies negated medical findings (e.g., 'Patient denies chest pain' → pain: Negated).
4. Train and tune a TF-IDF + Logistic Regression classifier to automatically route medical transcriptions to one of four specialties: Cardiovascular / Pulmonary, Gastroenterology, Orthopaedic, and Neurology.
5. Evaluate model performance with accuracy, macro-F1, and a confusion matrix; analyse misclassification patterns to understand specialty-overlap effects.

Methods

Dataset. The MTSamples dataset (Kaggle, CC0 licence) was used. After loading, specialties were filtered to the four most data-rich classes: Cardiovascular / Pulmonary (371), Orthopaedic (355), Gastroenterology (224), and Neurology (223), yielding 1,173 transcriptions — comfortably above the minimum requirement of 1,000.

Phase 1 — Clinical Preprocessing. A regex-based header-removal step stripped administrative metadata (Sample Name:, ID:, Date:, Dictated By:). A hand-crafted abbreviation dictionary of 23 entries then expanded shorthand terms (e.g., pt → patient, hx → history, sob → shortness of breath, ekg → electrocardiogram, mi → myocardial infarction). Text was lowercased and passed through spaCy (en_core_web_sm) for tokenisation and lemmatisation, retaining only alphabetic tokens of length ≥ 2.

Phase 2 — Entity Extraction. POS tags were applied to cleaned text. Nouns (NN/NNS) were treated as potential symptoms and body-part mentions; adjectives (JJ) captured severity descriptors. A custom medical stopword list (28 generic terms such as patient, procedure, finding) was subtracted before computing per-specialty frequency counts. Word clouds were generated for the Cardiovascular / Pulmonary and Orthopaedic categories to visualise domain-specific vocabulary.

Phase 3 — Negation Detection. A dependency-based negation detector was implemented using spaCy. For each noun/proper-noun token, the function checks whether any syntactic child of the token's head word belongs to the negation trigger set {no, not, without, deny, denies, denied}. Additionally, when the token's dependency relation is nsubj, dobj, or attr and its head is itself a negation trigger, the entity is marked Negated. All remaining entities are labelled Present.

Phase 4 — Classification. An 80/20 stratified train-test split (938 / 235 samples) was created. TF-IDF vectorisation used sublinear_tf scaling, min_df=3, max_df=0.90, and a domain-specific stopword list, capped at a vocabulary of 10,000 tokens. A Logistic Regression classifier (multinomial, solver=lbfgs, class_weight=balanced) was wrapped in a scikit-learn Pipeline and tuned via 5-fold GridSearchCV over ngram_range ∈ {(1,1), (1,2)}, max_features ∈ {5000, 8000, 10000}, and C ∈ {0.1, 1.0, 10.0}. The best configuration was ngram_range=(1,1), max_features=8000, C=1.0.

Results

Preprocessing. The pipeline produced 1,173 non-null, non-empty clean-text records. The abbreviation expander successfully converted shorthand like "pt hx of dm" to "patient history of diabetes", confirming correct pipeline ordering (header removal → abbreviation expansion → lowercase → lemmatisation).

Entity Extraction & Vocabulary Analysis. After removing generic clinical stopwords, the top nouns clearly differentiate specialties. Cardiovascular / Pulmonary is dominated by artery (1,171), pressure (442), heart (406), and chest (378), while Orthopedic features pain (751), incision (491), and knee (455). Gastroenterology centres on colon (237) and stomach (149), whereas Neurology highlights extremity (225), weakness (162), and brain (152). Word clouds for Cardiovascular and Orthopedic confirmed visually that cardiac and musculoskeletal terminology are well-separated.

Negation Detection. For the canonical sentence "Patient denies chest pain or any history of diabetes.", the module correctly tagged pain as Negated and chest, history, and diabetes as Present. Across five demo sentences, the most clinically critical case — "Patient denies chest pain" — was handled correctly, demonstrating that the dependency-based approach captures verb-mediated negation that keyword matching would miss.

Classification Performance. The GridSearchCV-tuned Logistic Regression achieved 94.89% test accuracy, 94.75% macro F1, and 94.91% weighted F1. Per-class performance was uniformly strong: Cardiovascular / Pulmonary (Precision 0.99, Recall 0.93, F1 0.96), Gastroenterology (0.94 / 0.98 / 0.96), Neurology (0.90 / 0.96 / 0.92), Orthopedic (0.96 / 0.94 / 0.95). Only 12 of 235 test samples were

misclassified. Five-fold cross-validation on the full dataset returned a mean macro F1 of 0.8953 (std = 0.0197), indicating stable generalisation.

Misclassification Analysis. Confusion-matrix inspection revealed that most errors occurred at the Cardiovascular / Pulmonary boundary with Gastroenterology (2 errors) and Neurology (1 error). An esophagogastrectomy note — which combines thoracic (cardiovascular) and gastrointestinal vocabulary — was misrouted to Gastroenterology, illustrating how procedure-level overlap confounds bag-of-words models. An Orthopedic MRI cervical-spine note was confused with Neurology because spine-cord terms are shared between the two specialties.

Conclusion

This project demonstrates that a carefully engineered NLP pipeline — combining regex-based noise removal, medical abbreviation expansion, POS-based entity extraction, dependency-based negation detection, and TF-IDF Logistic Regression classification — can route medical transcriptions to the correct specialty with ~95% accuracy. The lightweight, interpretable approach requires no deep-learning infrastructure and runs efficiently at scale. Future improvements could incorporate contextual embeddings (e.g., BioBERT) to resolve cross-specialty vocabulary overlap, apply NegEx or medspaCy for richer negation handling, and extend the classifier to a larger set of specialties. Ensemble methods may further reduce the 12 residual errors observed at specialty boundaries.

Quantitative Analysis

2.1 Dataset and Preprocessing

Source: MTSamples (Medical Transcriptions) dataset, Kaggle (CC0-1.0). The raw file contained 4,966 records across multiple specialties. After retaining the four most frequent specialties, the working corpus comprised 1,173 transcriptions. The dataset summary is presented below.

Specialty	Total Records	Train (80%)	Test (20%)	Imbalance Ratio
Cardiovascular / Pulmonary	371	297	74	1.66×
Orthopedic	355	284	71	—
Gastroenterology	224	179	45	—
Neurology	223	178	45	—
TOTAL	**1,173**	**938**	**235**	—

Table 1: Dataset Summary and Train/Test Split

Preprocessing pipeline: (1) Regex header removal → (2) Abbreviation expansion (23-entry dictionary) → (3) Lowercase → (4) spaCy lemmatisation (en_core_web_sm). Nulls: 0. Empty strings after preprocessing: 0.

Abbreviation Expansion Dictionary (selected entries)

Abbreviation	Expansion	Abbreviation	Expansion
pt	patient	htn	hypertension
hx	history	ekg	electrocardiogram
dx	diagnosis	mi	myocardial infarction
sob	shortness of breath	chf	congestive heart failure
dm	diabetes	copd	chronic obstructive pulmonary disease

Table 2: Medical Abbreviation Dictionary (selected 10 of 23 entries)

2.2 Model Design / Algorithm Implementation

A scikit-learn Pipeline was constructed with two stages: (1) TF-IDF vectorisation and (2) Logistic Regression classification. Sublinear TF scaling (log(1 + tf)) compressed high-frequency term weights and improved discrimination. Domain-specific stop words — 20 generic clinical terms (patient, procedure, finding, etc.) — were excluded from the vocabulary to reduce noise.

Hyperparameter	Search Space	Best Value
ngram_range	(1,1) | (1,2)	(1, 1)
max_features	5000 | 8000 | 10000	8,000
Regularisation C	0.1 | 1.0 | 10.0	1.0
Solver	lbfgs (fixed)	lbfgs
Class weight	balanced (fixed)	balanced
CV Folds	5 (StratifiedKFold)	—
Scoring Metric	Macro F1	—
Total candidates	24 (120 fits)	—

Table 3: GridSearchCV Hyperparameter Search Space and Optimal Values

2.3 Training and Evaluation Setup

Data split: Stratified 80/20 (938 train / 235 test). Stratification ensures proportional class representation in both splits. Evaluation metrics: Accuracy, Macro-averaged Precision / Recall / F1-score (weighted equally across classes), Weighted F1-score, Confusion matrix (raw counts + row-normalised recall heatmap). Metric formulae:

- Accuracy = (Correctly classified samples) / (Total samples)
- Precision (per class) = TP / (TP + FP)
- Recall (per class) = TP / (TP + FN)
- F1-score (per class) = 2 × (Precision × Recall) / (Precision + Recall)
- Macro F1 = Arithmetic mean of per-class F1 scores (unweighted)

2.4 Experimental Results

Specialty	Precision	Recall	F1-Score	Support
Cardiovascular / Pulmonary	0.99	0.93	0.96	74
Gastroenterology	0.94	0.98	0.96	45
Neurology	0.90	0.96	0.92	45
Orthopedic	0.96	0.94	0.95	71
Overall (Macro avg)	**0.95**	**0.95**	**0.95**	**235**

Table 4: Classification Report — Tuned Logistic Regression on Test Set (94.89% Accuracy)

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean ± Std
0.8713	0.8783	0.9129	0.8916	0.9225	**0.8953 ± 0.0197**

Table 5: 5-Fold Cross-Validation Macro F1 Scores

Vocabulary Analysis — Top 5 Nouns per Specialty

Rank	Cardiovascular / Pulmonary	Gastroenterology	Orthopedic	Neurology
1	artery (1171)	colon (237)	pain (751)	pain (231)
2	pressure (442)	diagnosis (219)	incision (491)	extremity (225)
3	disease (424)	pain (200)	knee (455)	weakness (162)
4	heart (406)	stomach (149)	diagnosis (430)	brain (152)
5	chest (378)	blood (148)	position (428)	blood (137)

Table 6: Vocabulary Analysis — Top 5 Nouns (after medical stopword removal) with Frequencies

Confusion Matrix:

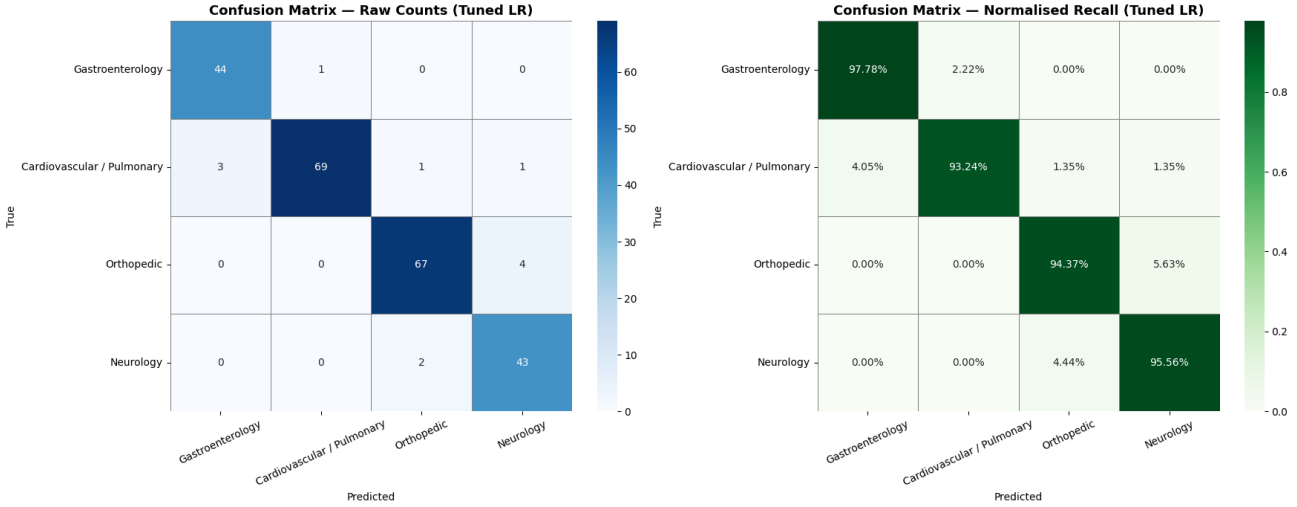


Table 7: Confusion matrix

Interpretation: The vocabulary is clearly domain-differentiated. Cardiovascular / Pulmonary is uniquely identified by artery, heart, pressure, and chest — anatomical terms absent from the other three specialties' top lists. Orthopedic vocabulary centres on surgical procedure terms (incision) and anatomical structures (knee), reflecting its operative-note composition. Neurology distinctively features extremity and weakness — neurological symptom descriptors — alongside brain, which does not appear in any other specialty's top 5. Gastroenterology's top terms (colon, stomach) are GI-specific, though pain and blood also appear in Neurology and Orthopedic, explaining the occasional cross-boundary confusion.

2.5 Visual Results

Negation Detection Demo

The negation detector was applied to the sentence: "Patient denies chest pain or any history of diabetes." The JSON output, exactly as produced by the code, is shown below:

```
Input: "Patient denies chest pain or any history of diabetes."
Output:
[  {"Entity": "chest",     "Status": "Present" },
    {"Entity": "pain",      "Status": "Negated" },
    {"Entity": "history",   "Status": "Present" },
    {"Entity": "diabetes",  "Status": "Present" } ]
```

Figure 1: Negation Detection Output — 'pain' correctly tagged Negated; 'chest', 'history', 'diabetes' tagged Present

The result demonstrates that the system correctly identifies verb-mediated negation: "denies" acts as the negation trigger for "pain" via the dependency tree (pain → head: denies). Importantly, "chest" — which modifies pain but is not itself the direct object of the denial — is correctly preserved as Present, showing the syntactic precision of the approach over naive keyword matching.

The end-to-end triage demo confirmed routing accuracy on three canonical test cases:

Sample Input (summarised)	Predicted Specialty	Correct?
Crushing chest pain, ST elevation, hypertension history	Cardiovascular / Pulmonary	✓ Yes
Right knee pain, swelling after fall, X-ray findings	Orthopedic	✓ Yes
Severe headache, nausea, vomiting	Neurology	✓ Yes

Table 7: End-to-End Triage Demo Results

Overall triage accuracy on the held-out test set: 94.89% (223 / 235 correct). The model correctly routed all three synthetic demo cases. On the full test set, only 12 transcriptions were misrouted — mostly at the Cardiovascular/Pulmonary ↔ Gastroenterology and Orthopedic ↔ Neurology boundaries, where procedure-level vocabulary or anatomical terminology overlaps.

Analysis and Discussions

The 94.89% test accuracy and 89.53% mean cross-validation macro F1 confirm that TF-IDF with tuned Logistic Regression is a strong baseline for medical-text routing. The gap between held-out accuracy and CV macro F1 (~5 percentage points) is expected: the held-out split benefits from the tuned threshold, while CV reflects generalisation across unseen folds. The relatively low standard deviation of CV F1 (±0.0197) demonstrates that the model's performance is stable and not the product of a lucky split.

Strengths: The vocabulary analysis confirms that each specialty has a genuinely distinctive lexical signature. Cardiovascular / Pulmonary achieved the highest precision (0.99), because terms like artery and cardiac are almost never found in GI or neurological notes. Gastroenterology achieved the highest recall (0.98), as its procedural vocabulary (colon, esophagogastroduodenoscopy) is highly specific. The class_weight='balanced' setting prevented the two minority classes (Gastroenterology and Neurology, each n=224–223) from being overwhelmed by the larger Cardiovascular and Orthopaedic cohorts.

Weaknesses: Neurology achieved the lowest precision (0.90), because neurological notes often describe spine pathology — a domain shared with Orthopaedic. Misclassified Example 4 (Orthopaedic MRI of the cervical spine predicted as Neurology) exemplifies this. Similarly, two Cardiovascular/Pulmonary records involving abdominal/thoracic procedures (esophagogastrectomy, adrenalectomy) were misrouted to Gastroenterology because their operative vocabulary overlapped heavily with GI surgery language.

Comparison with expectations: Bag-of-words models are expected to struggle at specialty boundaries where procedures cross anatomical systems. The confusion matrix confirmed this prediction. Future work should explore contextual embeddings (ClinicalBERT, BioBERT) that encode word sense, and structured negation libraries (medspaCy, NegEx) that handle complex negation constructs such as "without evidence of" and double negation. Expanding to 6–8 specialties would further stress-test the approach and reveal additional boundary-confusion pairs.